

$$\begin{array}{ccccc}
 & & s_{X,Y} & & \\
 & \longleftarrow & & \longrightarrow & \\
 T[TX, Y] & \xrightarrow{T[TX, \eta_Y]} & & & T[TX, TY] \\
 \eta_{[TX, Y]} \uparrow & & \eta \text{ nat} & & \nearrow \eta_{[TX, TY]} \\
 [TX, Y] & \xrightarrow{[TX, \eta_Y]} & [TX, TY] & & \\
 [\eta_X, \eta_Y] \downarrow & & \text{\textit{T}-alg} & & \downarrow a_{[TX, TY]} \\
 & [-, =] \text{ functor} & & & \\
 [X, TY] & \xleftarrow{[\eta_X, TY]} & & & [TX, TY]
 \end{array}$$